

AP CSP | Lesson 26: Parallel & Distributed

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 45 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Q5 requires exactly TWO choices. Show paths, calculations and reasons. Treat all networks and timings as the stated classroom models.

1. Which describes sequential computing? [1 mark]

- A. All operations occur at once. B. Multiple devices are required.
C. Operations occur one at a time in order. D. Communication has no cost.

Answer: _____

2. Which statement correctly relates parallel and distributed computing? [1 mark]

- A. They can occur together in one solution. B. They are always mutually exclusive.
C. Parallel work always requires multiple devices. D. Distributed work has no sequential portion.

Answer: _____

3. Setup costs 2. Jobs [6,4,3,3] use two processors, list-order assignment to earliest free processor, lower index on ties. No overhead. Runtime? [1 mark]

- A. 8 B. 9 C. 10 D. 11

Answer: _____

4. Sequential runtime is 18; parallel runtime is 8. What is speedup? [1 mark]

- A. 0.44 B. 2.25 C. 10 D. 26

Answer: _____

5. Which TWO statements are correct? [1 mark]

- A. Required serial work limits speedup. B. More processors always halve time.
C. More processors replace missing input data. D. Coordination overhead can reduce a parallel advantage.

Answer: _____

6. Separate model: setup 2, two independent 1-unit jobs. Sequential total 4. With two processors and 2 extra overhead units, what is parallel total? [1 mark]

- A. 2 B. 3 C. 5 D. 8

Answer: _____

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Written responses | Questions 7-9

7. [4 marks]

The fixed model has setup 2 and jobs A=6, B=4, C=3, D=3. (a) Sum the job work. (b) Give one-processor runtime. (c) Explain why setup cannot overlap jobs. (d) State a condition needed to overlap the jobs.

8. [4 marks]

Use Q7's model with 2 processors. Assign jobs in order to the earliest free processor, lower index on ties. (a) Which processor receives C and when does C start? (b) Which receives D and when? (c) When does P1 finish? (d) Give total runtime. Times start at program start, including setup.

9. [4 marks]

Use Q7's model with 4 processors. (a) Give zero-overhead runtime. (b) Calculate speedup from 18. (c) Add 6 overhead units after jobs: give runtime. (d) Why can extra processors beyond 4 not reduce the zero-overhead time here?

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Written responses | Questions 10-12

10. [4 marks]

A task has sequential runtime 18 and 2 unavoidable serial units. (a) State a lower bound on parallel runtime. (b) Give the corresponding upper bound on speedup. (c) Does the bound guarantee it is achievable? Explain. (d) Name a further limit.

11. [4 marks]

A distributed job needs all four packet chunks before processing. Only IDs 1,2,4 arrive. (a) Explain why input is incomplete. (b) Give a recovery action. (c) Explain why extra processors do not fix this. (d) State what makes the computing distributed.

12. [4 marks]

A design has an alternate network path and uses four processors. (a) State a fault it may tolerate. (b) State a network limit. (c) State a computing performance limit. (d) Explain why “always reliable and four times faster” is unjustified.
